WORKED SOLUTION

STEP III 2019, Question 5

Pure • Integration by Substitution

STEP III 2019, Question 5

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

The function in (i) maps $[0,\infty)$ smoothly and increasingly onto $[0,1)$. So scaling it by a constant turns a y-integral over part of $[0,c)$ into an x-integral over part of $[0,\infty)$. It also makes c^2-y^2 a perfect square in disguise, which removes the awkward square root. Once the root has gone we're left with a rational integrand, and we can do that with the arctangent integral given in the note. The reciprocal substitution $y=1/u$ links the later integrals back to the first one.

Solution

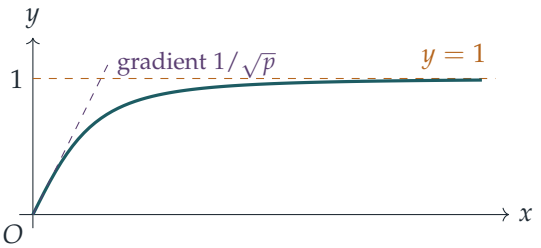
Part (i)

With $f(x)=x(x^2+p)^{-1/2}$ and $p>0$, we have $f(0)=0$ and

$$f'(x)=(x^2+p)^{-1/2}-x^2(x^2+p)^{-3/2}=\frac{p}{(x^2+p)^{3/2}}>0,$$

$$f''(x)=-\frac{3px}{(x^2+p)^{5/2}}<0 \quad \text{for } x>0.$$

So the curve leaves the origin with gradient $1/\sqrt{p}$ and rises steadily, bending downwards (it's concave). Since $f(x)=1/\sqrt{1+p/x^2}$ for $x>0$, we get $f(x)\to 1$ from below as $x\to\infty$, so $y=1$ is an asymptote that the curve never reaches.



Part (ii)

As printed, the substitution has b in the numerator. That doesn't work: it makes $b^2 - y^2$ neat but leaves the square root of $c^2(x^2+p) - b^2x^2$, which isn't a perfect square. The substitution that does work (and the one the examiners intended) has c in the numerator:

$$y = \frac{cx}{\sqrt{x^2+p}}, \qquad p > 0.$$

By part (i), this is an increasing map from $x \in [0, \infty)$ onto $y \in [0, c)$, which is the range where $\sqrt{c^2 - y^2}$ is defined. Now

$$\begin{aligned} c^2 - y^2 &= \frac{c^2(x^2+p) - c^2x^2}{x^2+p} = \frac{c^2 p}{x^2+p}, \\ b^2 - y^2 &= \frac{b^2 p + (b^2 - c^2)x^2}{x^2+p}, \\ \frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{cp}{(x^2+p)^{3/2}} \quad \text{(from part (i)).} \end{aligned}$$

Since $c > 0$ and $p > 0$, $\sqrt{c^2 - y^2} = c\sqrt{p}/\sqrt{x^2+p}$. Substituting,

$$I = \int \frac{x^2+p}{b^2 p + (b^2-c^2)x^2} \cdot \frac{\sqrt{x^2+p}}{c\sqrt{p}} \cdot \frac{cp}{(x^2+p)^{3/2}} \,\mathrm{d}x = \int \frac{\sqrt{p}}{b^2 p + (b^2-c^2)x^2} \,\mathrm{d}x.$$

Choosing $p = 1$ gives

$$I = \int \frac{1}{b^2 + (b^2-c^2)x^2} \,\mathrm{d}x,$$

which is the required result. The substitution is $y = cx/\sqrt{x^2+1}$, and its inverse is $x = y/\sqrt{c^2 - y^2}$.

The first integral. Take $b = \sqrt{3}$ and $c = \sqrt{2}$. Then $x = y/\sqrt{2-y^2}$, so $y = 1$ gives $x = 1$, and $x \to \infty$ as $y \to \sqrt{2}$. The original integrand blows up at $y = \sqrt{2}$ like $(2-y^2)^{-1/2}$, which is integrable. After the substitution the integral is a convergent improper integral on $[1, \infty)$:

$$\begin{aligned} \int_1^{\sqrt{2}} \frac{\mathrm{d}y}{(3-y^2)\sqrt{2-y^2}} &= \int_1^{\infty} \frac{\mathrm{d}x}{3+x^2} = \frac{1}{\sqrt{3}} \Big[\tan^{-1} \frac{x}{\sqrt{3}} \Big]_1^{\infty} \\ &= \frac{1}{\sqrt{3}} \Big(\frac{\pi}{2} - \frac{\pi}{6} \Big) = \frac{\pi}{3\sqrt{3}}. \end{aligned}$$

Ask yourself The new integral has $3y^2 - 1$ and $2y^2 - 1$ where the old one had $3 - y^2$ and $2 - y^2$. What substitution swaps the roles of y^2 and the constant?

The second integral. Put $y = 1/u$ with $u > 0$, so that $\mathrm{d}y = -\mathrm{d}u/u^2$. Then

$$3y^2 - 1 = \frac{3 - u^2}{u^2}, \qquad \sqrt{2y^2 - 1} = \frac{\sqrt{2 - u^2}}{u},$$

and the limits $y = \frac{1}{\sqrt{2}}$ and $y = 1$ become $u = \sqrt{2}$ and $u = 1$. So

$$\int_{1/\sqrt{2}}^{1} \frac{y\,\mathrm{d}y}{(3y^2 - 1)\sqrt{2y^2 - 1}} = \int_{\sqrt{2}}^{1} \frac{1}{u} \cdot \frac{u^2}{3 - u^2} \cdot \frac{u}{\sqrt{2 - u^2}} \cdot \left(-\frac{\mathrm{d}u}{u^2}\right) = \int_{1}^{\sqrt{2}} \frac{\mathrm{d}u}{(3 - u^2)\sqrt{2 - u^2}}.$$

This is exactly the first integral, so its value is also $\dfrac{\pi}{3\sqrt{3}}$.

Part (iii)

The same reciprocal substitution $y = 1/u$ turns the integrand into $u/\big((3 - u^2)\sqrt{2 - u^2}\big)$. The extra factor of u now invites $w = \sqrt{2 - u^2}$, since then $3 - u^2 = 1 + w^2$. Combining the two steps gives a single substitution:

$$w = \frac{\sqrt{2y^2 - 1}}{y} = \sqrt{2 - \frac{1}{y^2}}, \qquad \text{so} \qquad w^2 = 2 - \frac{1}{y^2}, \quad 2w\,\mathrm{d}w = \frac{2}{y^3}\,\mathrm{d}y.$$

For $y \in [\frac{1}{\sqrt{2}}, 1]$, w increases from 0 to 1. Also $\mathrm{d}y = wy^3\,\mathrm{d}w$, $\sqrt{2y^2 - 1} = wy$, and $3y^2 - 1 = y^2\big(3 - \frac{1}{y^2}\big) = y^2(1 + w^2)$. Hence

$$\int_{1/\sqrt{2}}^{1} \frac{\mathrm{d}y}{(3y^2 - 1)\sqrt{2y^2 - 1}} = \int_0^1 \frac{wy^3\,\mathrm{d}w}{y^2(1 + w^2) \cdot wy} = \int_0^1 \frac{\mathrm{d}w}{1 + w^2} = \frac{\pi}{4}.$$

(The singularity at $y = 1/\sqrt{2}$ is again only of the integrable type $(2y^2 - 1)^{-1/2}$, and the substitution removes it.)

This is the part (ii) method in disguise. Writing $x = 1/w$ gives $y = x/\sqrt{2(x^2 - 1)}$, which is the general substitution with $c^2 = \frac{1}{2}$ and $p = -1$. That's why part (i) allowed p to be negative.

Checks

- $\pi/(3\sqrt{3}) \approx 0.6046$ and $\pi/4 \approx 0.7854$. Both values were confirmed by numerical integration of the original integrals, after a trigonometric substitution to smooth out the endpoint singularities.
- Rough size check for the first integral: on $[1, \sqrt{2}]$ we have $1 \le 3 - y^2 \le 2$, so the integral lies between $\frac{1}{2}$ and 1 times $\int_1^{\sqrt{2}} (2 - y^2)^{-1/2}\,\mathrm{d}y = \frac{\pi}{4}$. That gives a range of 0.39 to 0.79, and 0.6046 sits comfortably inside it.
- Check the part (ii) identity at one point: with $b = \sqrt{3}$, $c = \sqrt{2}$ and $x = 1$ we get $y = 1$. The integrand is $\frac{1}{2}$, $\mathrm{d}y/\mathrm{d}x = \sqrt{2}/2^{3/2} = \frac{1}{2}$, and their product is $\frac{1}{4} = 1/(3 + 1)$, as expected.

Common slips

- Persisting with the printed b in the substitution and getting stuck: the square root you need to remove is $\sqrt{c^2 - y^2}$, so c must be the scale factor.
- Converting the limits wrongly. Here $y = \sqrt{2} = c$ corresponds to $x = \infty$, not to a finite value of x.
- Dropping the minus sign from $\mathrm{d}y = -\mathrm{d}u/u^2$, or forgetting to swap the limits back, which gives a negative answer for a positive integrand.

Beyond STEP

The substitutions here are rationalising substitutions. They turn an integrand that involves the square root of a quadratic into a rational function, and partial fractions or $\tan^{-1}$ then finish the job. In first-year analysis you meet the theory behind the improper integrals used here: when an integral with an infinite limit or an unbounded integrand converges, and why substitution is still valid for such integrals. Bartle and Sherbert's *Introduction to Real Analysis* covers both carefully.

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